

Functions of several variables.

[SUBMIT ANSWERS TO ALL QUESTIONS]

Total differential of $f(x, y, z)$: $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$.

Gradient vector: $\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k} \equiv \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$.

Directional derivative along unit vector $\mathbf{T}$: $\frac{\partial f}{\partial s} = \nabla f \cdot \mathbf{T} = \frac{\partial f}{\partial x}T_x + \frac{\partial f}{\partial y}T_y + \frac{\partial f}{\partial z}T_z$.

Taylor's expansion for a function $f(x, y)$:

$$f(a+h, b+k) = f(a, b) + f'_x h + f'_y k + \frac{1}{2} [f''_{xx} h^2 + 2f''_{xy} hk + f''_{yy} k^2] + \dots,$$

where the derivatives $f'_x \equiv \partial f / \partial x$, $f'_y \equiv \partial f / \partial y$, $f''_{xx} \equiv \partial^2 f / \partial x^2$, etc., are evaluated at (a, b) .

Stationary point: $f'_x(a, b) = 0$, $f'_y(a, b) = 0$. Let $A = f''_{xx}(a, b)$, $B = f''_{xy}(a, b)$, $C = f''_{yy}(a, b)$, and $\Delta = AC - B^2$. For $\Delta > 0$ this point is an extremum: $A > 0$ – minimum, $A < 0$ – maximum. For $\Delta < 0$ this is a saddle point. For $\Delta = 0$ further investigation is needed.

1. Find the partial derivatives of the following functions:

$$(a) z = \frac{x-y}{x+y}, \quad (b) z = \sqrt{x^2 - y^2}, \quad (c) z = \ln(x + \sqrt{x^2 + y^2}), \quad (d) z = \arctan \frac{y}{x}.$$

[20]

2. Find the gradient vectors of the following functions:

$$(a) f(x, y, z) = x^2 + y^2 + z^2 + xy + xz + yz, \quad (b) f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}.$$

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3. Find the derivative of the function $f(x, y) = x^3 - 2x^2y + xy^2 + 1$ at point $P(1, 2)$ in the direction from this point to point $Q(4, 6)$.

[20]

4. Find the second-order partial derivatives for $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, and verify that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}, \quad \frac{\partial^2 f}{\partial z \partial x} = \frac{\partial^2 f}{\partial x \partial z}, \quad \frac{\partial^2 f}{\partial z \partial y} = \frac{\partial^2 f}{\partial y \partial z}, \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

[20]

5. Find the stationary points of the following functions and use the second-order derivatives to determine (if possible), whether they are maxima, minima, or saddle points:

$$(a) z = x^2 + xy + y^2 - 2x - y, \quad (b) z = e^{-(x^2+y^2)}, \quad (c) z = x^3 + y^3 - 3xy,$$

$$(d) z = 2x^4 + y^4 - x^2 - 2y^2.$$

[Hint: in (d) you need to find and investigate all nine(!) stationary points.]

[20]

MTH1021 PROBLEM SHEET 18

SOLUTIONS.

FUNCTIONS OF SEVERAL VARIABLES.

Q1

$$(a) \quad z = \frac{x-y}{x+y}$$

$$\frac{\partial z}{\partial x} = \frac{1 \cdot (x+y) - (x-y) \cdot 1}{(x+y)^2} \quad (\text{Quotient Rule})$$

$$= \frac{2y}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(-1)(x+y) - (x-y) \cdot 1}{(x+y)^2}$$

$$= \frac{-2x}{(x+y)^2}$$

$$(b) \quad z = \sqrt{x^2 - y^2} = (x^2 - y^2)^{1/2}$$

$$\frac{\partial z}{\partial x} = \frac{1}{2} (x^2 - y^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 - y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2} (x^2 - y^2)^{-1/2} \cdot (-2y) = -\frac{y}{\sqrt{x^2 - y^2}}$$

$$(c) \quad z = \ln(x + \sqrt{x^2 + y^2})$$

$$\frac{\partial z}{\partial x} = \frac{1}{x + \sqrt{x^2 + y^2}} \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right)$$

$$= \frac{\sqrt{x^2 + y^2} + x}{\sqrt{x^2 + y^2} (x + \sqrt{x^2 + y^2})} = \frac{1}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{x + \sqrt{x^2 + y^2}} \cdot \left(\frac{y}{\sqrt{x^2 + y^2}} \right)$$

$$= \frac{y}{\sqrt{x^2 + y^2} (x + \sqrt{x^2 + y^2})}$$

(d) $z = \arctan(y/x)$

$$\frac{\partial z}{\partial x} = \left(\frac{1}{1 + (y/x)^2} \right) \cdot \left(-\frac{y}{x^2} \right) = -\frac{y}{x^2 + y^2}$$

$$\frac{\partial z}{\partial y} = \left(\frac{1}{1 + (y/x)^2} \right) \cdot \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2}$$

Q2.

(a) $f(x, y, z) = x^2 + y^2 + z^2 + xy + xz + yz$

$$\frac{\partial f}{\partial x} = 2x + y + z$$

$$\frac{\partial f}{\partial y} = 2y + x + z$$

$$\frac{\partial f}{\partial z} = 2z + x + y$$

Hence the gradient vector is

$$\underline{\nabla} f = (2x + y + z, 2y + x + z, 2z + x + y)$$

or

$$\underline{\nabla} f = (2x + y + z) \underline{i} + (2y + x + z) \underline{j} + (2z + x + y) \underline{k}$$

$$(b) \quad f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

$$= (x^2 + y^2 + z^2)^{-1/2}.$$

$$\frac{\partial f}{\partial x} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x = -\frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial f}{\partial y} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2y = -\frac{y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial f}{\partial z} = -\frac{z}{(x^2 + y^2 + z^2)^{3/2}}.$$

Hence

$$\underline{\nabla} f = \left(\frac{-x}{(x^2 + y^2 + z^2)^{3/2}}, \frac{-y}{(x^2 + y^2 + z^2)^{3/2}}, \frac{-z}{(x^2 + y^2 + z^2)^{3/2}} \right)$$

Note

$$\text{If } \underline{r} = x \underline{i} + y \underline{j} + z \underline{k} = (x, y, z)$$

$$|\underline{r}| = \sqrt{x^2 + y^2 + z^2} = r$$

and

$$\underline{\nabla} f = -\frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= -\frac{(x \underline{i} + y \underline{j} + z \underline{k})}{(x^2 + y^2 + z^2)^{3/2}}.$$

We have shown that.

$$\underline{\nabla} \left(\frac{1}{r} \right) = -\frac{\underline{r}}{r^3}.$$

Q3.

$$F(x, y) = x^3 - 2x^2y + xy^2 + 1$$

In order to find a directional derivative, we first need to find the gradient vector

$$\frac{\partial F}{\partial x} = 3x^2 - 4xy + y^2$$

$$\frac{\partial F}{\partial y} = -2x^2 + 2xy$$

$$\Rightarrow \nabla F = (3x^2 - 4xy + y^2, -2x^2 + 2xy)$$

At point $P(1, 2)$ $x=1$ $y=2$ and

$$\nabla F = (3 - 8 + 4, -2 + 4)$$

$$\nabla F = (-1, 2) = -\underline{i} + 2\underline{j}$$

The direction from point P to point $Q(4, 6)$ is given by the vector $\overrightarrow{PQ} = (4-1, 6-2) = (3, 4)$.

The unit vector in the direction of $\overrightarrow{PQ}$ is

$$\underline{T} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{(3, 4)}{\sqrt{3^2 + 4^2}} = \frac{(3, 4)}{5} = (3/5, 4/5)$$

$$\underline{T} = 3/5 \underline{i} + 4/5 \underline{j}$$

The directional derivative of $F(x, y)$ at point $(1, 2)$ in the direction of $\underline{T}$ is given by

$$\begin{aligned} \frac{\partial F}{\partial s} &= \nabla F \cdot \underline{T} = (-\underline{i} + 2\underline{j}) \cdot (3/5 \underline{i} + 4/5 \underline{j}) \\ &= -3/5 + 8/5 \\ &= 5/5 \\ &= 1 \end{aligned}$$

$$Q4. \quad f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}.$$

$$\frac{\partial f}{\partial x} = -\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \quad (\text{see question 3}).$$

$$\frac{\partial f}{\partial y} = -\frac{y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial f}{\partial z} = -\frac{z}{(x^2 + y^2 + z^2)^{3/2}}.$$

Second order partial derivatives

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(-\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) \quad \text{Using the quotient rule} \\ &= - \left[\frac{(x^2 + y^2 + z^2)^{3/2} - x \cdot \frac{3}{2} (x^2 + y^2 + z^2)^{1/2} \cdot 2x}{(x^2 + y^2 + z^2)^3} \right] \\ &= - \left[\frac{x^2 + y^2 + z^2 - 3x^2}{(x^2 + y^2 + z^2)^{5/2}} \right] = \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}}. \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(-\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) \\ &= \frac{3}{2} \frac{x \cdot 2y}{(x^2 + y^2 + z^2)^{5/2}} = \frac{3xy}{(x^2 + y^2 + z^2)^{5/2}}. \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial z \partial x} &= \frac{\partial}{\partial z} \left(-\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) \\ &= + \frac{3}{2} \frac{x \cdot 2z}{(x^2 + y^2 + z^2)^{5/2}} = \frac{3xz}{(x^2 + y^2 + z^2)^{5/2}}. \end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{-y}{(x^2+y^2+z^2)^{3/2}} \right) \\ &= + \frac{3}{2} \frac{y \cdot 2x}{(x^2+y^2+z^2)^{5/2}} = \frac{3xy}{(x^2+y^2+z^2)^{5/2}}\end{aligned}$$

$$= \frac{\partial^2 f}{\partial y \partial x}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial z} &= \frac{\partial}{\partial x} \left(\frac{-z}{(x^2+y^2+z^2)^{3/2}} \right) \\ &= + \frac{3}{2} \frac{z \cdot 2x}{(x^2+y^2+z^2)^{5/2}} = \frac{3xz}{(x^2+y^2+z^2)^{5/2}}\end{aligned}$$

$$= \frac{\partial^2 f}{\partial z \partial x}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(- \frac{y}{(x^2+y^2+z^2)^{3/2}} \right) \\ &= - \left[\frac{(x^2+y^2+z^2)^{3/2} - y \cdot \frac{3}{2} (x^2+y^2+z^2)^{1/2} \cdot 2y}{(x^2+y^2+z^2)^3} \right] \\ &= - \left[\frac{x^2+y^2+z^2 - 3y^2}{(x^2+y^2+z^2)^{5/2}} \right] = \frac{2y^2 - x^2 - z^2}{(x^2+y^2+z^2)^{5/2}}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial z \partial y} &= \frac{\partial}{\partial z} \left(- \frac{y}{(x^2+y^2+z^2)^{3/2}} \right) \\ &= + \frac{3}{2} \frac{y \cdot 2z}{(x^2+y^2+z^2)^{5/2}} = \frac{3yz}{(x^2+y^2+z^2)^{5/2}}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y \partial z} &= \frac{\partial}{\partial y} \left(\frac{-z}{(x^2+y^2+z^2)^{3/2}} \right) \\ &= \frac{3}{2} \left(\frac{z \cdot 2y}{(x^2+y^2+z^2)^{5/2}} \right) = \frac{3zy}{(x^2+y^2+z^2)^{5/2}} = \frac{\partial^2 f}{\partial z \partial y}\end{aligned}$$

Finally

$$\frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z} \left(\frac{-z}{(x^2 + y^2 + z^2)^{3/2}} \right)$$

$$= \frac{2z^2 - x^2 - y^2}{(x^2 + y^2 + z^2)^{5/2}}$$

(similar to $\frac{\partial^2 f}{\partial x^2}$
and $\frac{\partial^2 f}{\partial y^2}$)

We observe that the mixed partial derivatives are the same, independent of the order of differentiation

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}, \quad \frac{\partial^2 f}{\partial z \partial x} = \frac{\partial^2 f}{\partial x \partial z}, \quad \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 f}{\partial z \partial y}$$

Now

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$+ \frac{2y^2 - x^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{2z^2 - x^2 - y^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$= \frac{\cancel{2x^2} - \cancel{y^2} - \cancel{z^2} + \cancel{2y^2} - \cancel{x^2} - \cancel{z^2} + \cancel{2z^2} - \cancel{x^2} - \cancel{y^2}}{(x^2 + y^2 + z^2)^{5/2}}$$

$$= 0 \quad \text{as required}$$

Q5.

(a) $z = x^2 + xy + y^2 - 2x - y$

$$\frac{\partial z}{\partial x} = 2x + y - 2 = 0 \quad \text{at stationary points} \dots (1)$$

$$\frac{\partial z}{\partial y} = x + 2y - 1 = 0 \dots (2)$$

Multiply (2) by 2 and subtract from (1).

$$y - 4y = 0 \Rightarrow y = 0$$

Substituting back into (2) gives $x = 1$

Hence $(1, 0)$ is a stationary point.

To determine its nature we need to find the second order derivatives

$$\frac{\partial^2 z}{\partial x^2} = 2 = A, \quad \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} (2x + y - 2) = 1 = B$$

$$\frac{\partial^2 z}{\partial y^2} = 2 = C$$

$$\Delta = AC - B^2 = (2)(2) - 1^2 = 3 > 0$$

and $A > 0$

hence $(1, 0)$ is a minimum with

$$z = 1 - 2 = -1 \quad \text{at this point.}$$

(b) $z = e^{-(x^2 + y^2)}$

$$\frac{\partial z}{\partial x} = -2x e^{-(x^2 + y^2)} = 0$$

$$\frac{\partial z}{\partial y} = -2y e^{-(x^2 + y^2)} = 0$$

These equations can only be satisfied at $x = 0, y = 0$. Hence $(0, 0)$ is the only critical point.

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= \frac{d}{dx} (-2x e^{-(x^2+y^2)}) \\ &= -2e^{-(x^2+y^2)} + 4x^2 e^{-(x^2+y^2)} \\ &= 2(2x^2 - 1) e^{-(x^2+y^2)} = A.\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial^2 z}{\partial y \partial x} = \frac{d}{dy} (-2x e^{-(x^2+y^2)}) \\ &= 4xy e^{-(x^2+y^2)} = B.\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 z}{\partial y^2} &= \frac{d}{dy} (-2y e^{-(x^2+y^2)}) \\ &= 2(2y^2 - 1) e^{-(x^2+y^2)} = C.\end{aligned}$$

At point $(0,0)$

$$A = -2, \quad B = 0, \quad C = -2.$$

$$\Delta = AC - B^2 = 4 > 0$$

and since $A = -2 < 0$ this point is a
maximum At $(0,0)$ $z = e^{-0} = 1.$

$$(c) \quad z = x^3 + y^3 - 3xy$$

$$\frac{\partial z}{\partial x} = 3x^2 - 3y = 0$$

$$\frac{\partial z}{\partial y} = 3y^2 - 3x = 0$$

at stationary points

$$\therefore x^2 - y = 0 \quad \dots (1)$$

$$y^2 - x = 0 \quad \dots (2)$$

From (1) $y = x^2$ and substituting into (2)
 gives $x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$

This equation has two solutions
 $x=0$ and $x^3-1=0 \Leftrightarrow x=1$. The
 corresponding values of y are $y=0$ and $y=1$.
 Hence we have two stationary points
 $(0, 0)$ and $(1, 1)$

$$\frac{\partial^2 z}{\partial x^2} = 6x \quad ; \quad \frac{\partial^2 z}{\partial x \partial y} = -3 \quad ; \quad \frac{\partial^2 z}{\partial y^2} = 6y$$

$\parallel$ $\parallel$ $\parallel$
 A B C

At $(0, 0)$ $A=0$, $B=-3$, $C=0$

$\Delta = AC - B^2 = -9 < 0 \Rightarrow$ this is not
 an extremum but a saddle point.

At $(1, 1)$ $A=6$, $B=-3$, $C=6$.

$\Delta = AC - B^2 = 36 - 9 = 27 > 0$

Also $A > 0$ so this point is a minimum.
 The value of the function at this point is
 $z = 1 + 1 - 3 = -1$.

(d) $z = 2x^4 + y^4 - x^2 - 2y^2$

$\frac{\partial z}{\partial x} = 8x^3 - 2x = 0 \Rightarrow (4x^2 - 1)x = 0 \dots (1)$

$\frac{\partial z}{\partial y} = 4y^3 - 4y = 0 \Rightarrow y(y^2 - 1) = 0 \dots (2)$

From (1) $x=0$ or $x = \pm \frac{1}{2}$.

From (2) $y=0$ or $y = \pm 1$.

Hence we have nine stationary points.

$(0, 0)$, $(0, \pm 1)$, $(\pm \frac{1}{2}, 0)$, $(\pm \frac{1}{2}, \pm 1)$
 (two points) (two points) (four points)

$$\frac{\partial^2 z}{\partial x^2} = 24x^2 - 2 = A$$

$$\frac{\partial^2 z}{\partial x \partial y} = 0 = B$$

$$\frac{\partial^2 z}{\partial y^2} = 12y^2 - 4 = C.$$

At $(0, 0)$

$A = -2$, $C = -4$, $\Delta = AC - B^2 = 8 > 0$
 and since $A < 0$, this is a
maximum with $z = 0$.

At $(0, \pm 1)$

$A = -2$, $C = 8$, $\Delta = -16 < 0$ not an
 extremum - saddle points.

At $(\pm \frac{1}{2}, 0)$

$A = 4$, $C = -4$, $\Delta = -16 < 0$ not an
 extremum - saddle points.

At $(\pm \frac{1}{2}, \pm 1)$

$A = 4$, $C = 8$, $\Delta = 32 > 0$ and
 since $A > 0$ these four points are
minima, with the function

$$z = 2\left(\frac{1}{2}\right)^4 + 1^4 - \frac{1}{4} - 2$$

$$= \frac{1}{8} - 1 - \frac{1}{4}$$

$$= -1\frac{1}{8}.$$
